

FADA N'GOURMA, Burkina Faso

WMO: 655070

Lat: 12.033N

Lon: 0.367E

Elev: 1014

StdP: 14.17

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	63.0	64.9	19.3	15.0	85.0	22.6	17.6	83.3	14.3	78.5	13.1	81.7	3.0	40	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	20.2	105.0	69.7	103.5	70.0	102.1	70.0	80.7	90.4	80.0	89.9	79.3	89.5	5.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.2	152.3	85.7	77.4	148.2	84.8	76.9	145.6	84.3	45.2	90.4	44.4	90.1	43.8	89.6	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.5	11.6	10.0	DB	58.7	108.5	2.3	0.6	57.0	108.9	55.6	109.3	54.3	109.6	52.7	110.0
			WB	44.6	82.1	1.3	0.9	43.7	82.7	43.0	83.3	42.2	83.8	41.3	84.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.0	78.3	83.9	90.5	92.1	89.3	85.2	81.0	79.5	81.0	84.1	83.8	79.2
	DBStd	5.46	3.82	4.26	2.79	3.36	3.97	3.66	3.03	2.34	2.80	2.22	2.35	3.23
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12403	877	949	1256	1263	1219	1055	962	915	930	1058	1013	906
	CDD65	6928	412	529	791	813	754	605	497	450	480	593	563	441
	CDH74	88021	4499	6881	12279	12810	11172	7836	4999	3912	4722	6956	7108	4847
	CDH80	48726	2250	3997	8049	8573	6887	4030	1845	1164	1805	3464	4174	2488
Wind	WSAvg	4.2	4.6	4.5	4.0	5.1	5.9	5.2	4.3	3.7	3.0	3.1	2.9	4.3
Precipitation	PrecAvg	32.4	0.0	0.0	0.2	1.1	2.7	4.5	7.2	9.1	5.9	1.4	0.0	0.0
	PrecMax	42.5	0.1	0.5	1.6	5.5	5.5	8.5	15.3	17.6	13.1	6.0	0.5	1.5
	PrecMin	21.8	0.0	0.0	0.0	0.0	0.4	1.1	3.5	4.4	0.0	0.0	0.0	0.0
	PrecStd	5.3	0.0	0.1	0.4	1.2	1.3	1.5	2.6	3.1	2.5	1.2	0.1	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.9	104.0	106.4	107.0	104.9	99.8	95.4	90.9	94.2	99.0	100.6	98.3
		MCWB	63.0	66.5	67.2	71.3	75.6	77.8	77.6	79.5	79.6	71.5	67.4	64.8
	2%	DB	96.0	101.5	104.8	105.3	102.5	97.6	92.2	89.0	92.1	97.1	99.3	96.1
		MCWB	61.8	65.6	67.7	70.6	76.4	77.7	77.8	78.8	79.6	73.3	67.5	63.9
	5%	DB	93.6	99.3	103.4	103.8	100.5	95.6	90.1	87.4	90.3	95.2	98.0	94.0
		MCWB	60.9	64.9	67.7	71.1	76.7	77.5	77.8	78.3	79.4	75.4	67.4	62.7
	10%	DB	91.1	96.8	102.0	102.1	98.4	93.6	88.0	85.9	88.5	93.2	96.5	91.5
		MCWB	60.0	63.8	67.3	71.7	76.9	77.3	77.5	77.9	79.1	76.4	67.3	61.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.1	74.9	77.0	79.1	80.3	80.4	80.0	80.9	81.8	81.3	76.9	72.6
		MCDB	89.8	94.2	94.7	97.2	94.5	92.6	89.5	89.0	90.0	91.6	91.5	85.2
	2%	WB	64.1	72.2	75.6	78.1	79.3	79.5	79.1	79.7	80.8	80.2	75.3	66.9
		MCDB	91.6	88.2	94.9	96.0	94.4	91.7	88.2	87.0	89.2	90.5	90.1	89.8
	5%	WB	62.4	68.8	74.4	77.2	78.8	78.8	78.5	78.9	80.0	79.5	73.7	64.9
		MCDB	90.9	93.5	94.8	94.8	94.4	91.0	87.2	85.8	88.2	89.7	88.1	90.1
	10%	WB	60.9	65.5	72.8	76.4	78.2	78.2	77.9	78.2	79.3	78.7	72.1	63.0
		MCDB	89.4	94.5	94.0	93.8	93.3	89.8	86.0	84.4	87.1	88.8	87.4	88.8
Mean Daily Temperature Range	5% DB	MDBR	25.6	25.6	24.0	20.2	17.6	15.7	13.2	12.2	14.3	19.4	27.6	26.4
		MCDBR	28.7	28.1	25.7	22.6	20.1	17.9	16.2	14.7	17.1	22.3	28.7	28.3
		MCWBR	12.1	11.3	9.8	7.8	4.9	4.6	5.2	6.0	6.9	6.7	9.0	10.9
	5% WB	MCDBR	28.4	25.2	21.4	19.1	18.1	16.3	14.4	14.0	16.2	18.4	24.0	27.7
		MCWBR	12.8	11.5	9.8	6.1	5.3	5.3	5.7	6.5	7.2	7.2	8.5	11.9
Clear-Sky Solar Irradiance	taub	0.604	0.669	0.763	0.736	0.682	0.636	0.572	0.547	0.547	0.618	0.562	0.546	
	taud	1.716	1.581	1.444	1.578	1.647	1.752	1.948	2.006	1.988	1.796	1.812	1.831	
	Ebn at Noon	225	217	201	206	213	220	236	246	247	226	234	237	
	Edh at Noon	74	87	102	89	81	72	60	57	58	69	66	64	
All-Sky Solar Radiation	RadAvg	1759	1852	1919	1971	1910	1803	1689	1620	1745	1867	1801	1729	
	RadStd	89	88	90	63	67	63	49	59	78	59	46	62	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page